Data Science Across Domains

From Methods and Models to Business Applications

Rodrigo Kang

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Welcome

Welcome to my portfolio:

Data Science Across Domains
From Methods and Models to Business Applications

This portfolio presents a curated collection of applied data science projects developed across different domains. Rather than a traditional theoretical text, it is organized as a series of case studies showing how modern data science methods are used to frame problems, build models, and generate practical insights.

Each chapter focuses on a specific project. Problems are clearly defined, modeling assumptions are made explicit, and solutions are developed using a combination of statistical reasoning, machine learning methods, and computational experimentation. Results are evaluated using appropriate metrics and baseline comparisons.

The emphasis throughout the portfolio is on applied problem solving. Mathematical concepts and technical details are introduced only when they help explain the modeling choices or the behavior of the algorithms being used.

Every project presented here corresponds to a fully implemented workflow. The associated computational artifacts—written in Python and R—are available alongside the text, allowing readers to explore the code, reproduce the results, and understand the implementation details behind each solution.

This portfolio reflects how data science is practiced in real environments: combining sound modeling principles, careful experimentation, and clear communication of results.

Many of the projects presented here also draw on my background in physics and quantitative modeling, where mathematical reasoning and computational experimentation naturally complement modern machine learning techniques.

Each chapter can be read independently as a standalone project, while together they illustrate a consistent approach to building and applying data science solutions across different domains.

Preface

This portfolio reflects an ongoing effort to organize my work in applied data science into a coherent and practice-oriented body of projects. Rather than presenting isolated experiments or purely theoretical discussions, it documents complete workflows in which problem formulation, modeling choices, implementation, and evaluation are treated as parts of the same analytical process.

Structuring this material in book form mirrors how I approach machine learning problems in professional and research environments. In practice, successful applications rarely depend on algorithmic novelty alone. They usually emerge from a clear definition of the problem, explicit modeling assumptions, careful experimentation, and thoughtful interpretation of results within the context of the domain.

For this reason, the chapters are organized around concrete case studies. Each project begins with a well-defined problem and progresses through model design, training strategy, evaluation methodology, and discussion of limitations. Concepts from statistics, optimization, probability, and numerical methods appear throughout the text, but only when they are necessary to support modeling decisions or clarify the behavior of the methods being used.

An equally important aspect of this portfolio is its deliberate scope. It does not attempt to catalogue every possible technique or follow trends for their own sake. Instead, it focuses on methods that I have implemented, tested, and evaluated in real problem settings. The emphasis is on depth, clarity, and technical responsibility rather than breadth.

All projects presented here correspond to fully implemented solutions. Computational artifacts written in Python and R accompany the text and are referenced throughout, allowing readers to explore the implementation and reproduce the results if desired.

Ultimately, this portfolio serves both as a record of completed work and as a foundation for continued development. The projects documented here represent practical steps in an ongoing process of learning, experimentation, and refinement. My intention is that the structure of this work reflects not only technical capability, but also a disciplined and pragmatic approach to applying data science methods in real-world contexts.

About

I am a physicist and data scientist with a background in applied mathematics, machine learning, and computational modeling. My work focuses on applying quantitative methods to real-world problems, combining mathematical reasoning with practical implementation.

My academic training in physics has strongly shaped the way I approach data science and machine learning. My research included work on **quantum optimal control theory**, with particular emphasis on algorithmic design based on *shortcuts to adiabaticity* and the study of computational complexity in physical systems. This background continues to influence my modeling philosophy: clear problem formulation, explicit assumptions, and careful evaluation of model behavior.

Professionally, I have applied machine learning, advanced analytics, and optimization techniques in data-intensive industries. Much of my work has been developed within the **energy sector**, where I have contributed to quantitative modeling, decision-support systems, and optimization frameworks for gas trading and operational planning.

In the retail sector, I have worked on projects involving customer segmentation, churn prediction, and recommendation systems, balancing theoretical rigor with practical constraints such as data quality, interpretability, and deployment considerations.

More recently, within Oil & Gas research and development environments, my work has focused on integrating physical modeling, statistical inference, and machine learning to address complex industrial problems. These projects include predictive modeling for pipeline corrosion, machine learning approaches to production optimization, and simulation-based frameworks for decision-making under uncertainty.

Across both academic and industrial settings, my perspective remains consistent: machine learning methods such as Deep Learning and Reinforcement Learning are powerful computational tools, but their effectiveness ultimately depends on disciplined modeling, transparent assumptions, and careful validation.

Outside my professional work, I enjoy playing the bass guitar and maintain a longstanding interest in theology.

How to Read This Portfolio

This portfolio is organized around applied projects rather than purely theoretical chapters. Each chapter presents a self-contained case study showing how data science methods are used to address a specific problem, from formulation and modeling to evaluation and interpretation.

Projects can be read independently. Readers interested in a particular topic can navigate directly to the corresponding chapter without needing to follow the entire sequence.

Each project typically follows the same structure:

1. **Problem definition**
   A clear description of the objective and the context in which the problem arises.

2. **Modeling approach**
   Selection and development of appropriate statistical or machine learning methods.

3. **Implementation**
   Practical aspects of the solution, including data preparation, model training, and computational considerations.

4. **Evaluation**
   Performance assessment using appropriate metrics and baseline comparisons.

5. **Discussion and limitations**
   Interpretation of results, assumptions behind the model, and potential areas for improvement.

Theoretical concepts appear only when they help clarify modeling decisions or explain the behavior of the methods used. The emphasis throughout the portfolio is on applied problem solving rather than abstract presentation of algorithms.

Each chapter is accompanied by computational artifacts implemented in `Python` and `R`, allowing readers to explore the code and reproduce the results if desired.

Readers approaching this portfolio from different backgrounds may find the following paths useful:

- **Technical readers** may prefer to focus on modeling choices, validation strategies, and implementation details.
- **Practitioners and decision-makers** may focus on problem formulation, interpretation of results, and practical implications.

Overall, this portfolio should be viewed as a collection of applied case studies illustrating how quantitative reasoning, machine learning methods, and careful experimentation combine to produce practical data science solutions.

Chapter 1

CAPM and Fama–French Models

Work in progress.

1.1 Introduction

Chapter 2

Asset Allocation with PCA and Hierarchical Clustering

Work in progress.

2.1 Introduction

Chapter 3

Credit Card Fraud Detection

Work in progress.

3.1 Introduction

Chapter 4

Customer Lifetime Value Prediction

Work in progress.

4.1 Introduction

Chapter 5

Marketing Campaign Optimization

Work in progress.

5.1 Introduction

Chapter 6

RFM Segmentation with K-Means

Work in progress.

6.1 Introduction

Chapter 7

Geochemical Fingerprint Deconvolution

Work in progress.

7.1 Introduction

Chapter 8

Pore–Throat Detection in Porous Media

Work in progress.

8.1 Introduction

Chapter 9

Climate Forecasting with Time Series

Work in progress.

9.1 Introduction

Chapter 10

Microbial Population Dynamics (Lotka–Volterra Model)

Work in progress.

10.1 Introduction

Chapter 11

Optimal Hedging of Industrial Gas Demand

Work in progress.

11.1 Introduction

Chapter 12

Knowledge Extraction from Technical Documents with LLMs

Work in progress.

12.1 Introduction

Chapter 13

Customer Sentiment and Topic Analysis

Work in progress.

13.1 Introduction

Chapter 14

Scientific Literature Mining with NLP and LLMs

Work in progress.

14.1 Introduction

References